Alliance University

Alliance School of Advanced Computing

Inventory Management and Sales Visualization System

Advanced Python Programming Micro-Project



Course Code: E1CSA348

Submitted by: Hithesh Reddy(1123)

Mahammad Asif(1151)

Kishwar Sahil(1176)

Course In-charge: Dr.CHETHAN J SHELKE

Problem Statement & Objectives

Problem Statement

Manual inventory management requires repeated stock updates, sales calculations and record checking. It can be difficult to quickly identify available stock and sales performance.



Stock updates



Sales calculations



Record checking

Objectives

- Develop a simple inventory management application.
- Store product and sales information using SQLite.
- Provide a user-friendly Tkinter GUI.
- Automatically update stock after sales.
- Analyze sales using Pandas.
- Visualize category-wise sales using Matplotlib.

Technology Stack & System Architecture

Python
Core programming

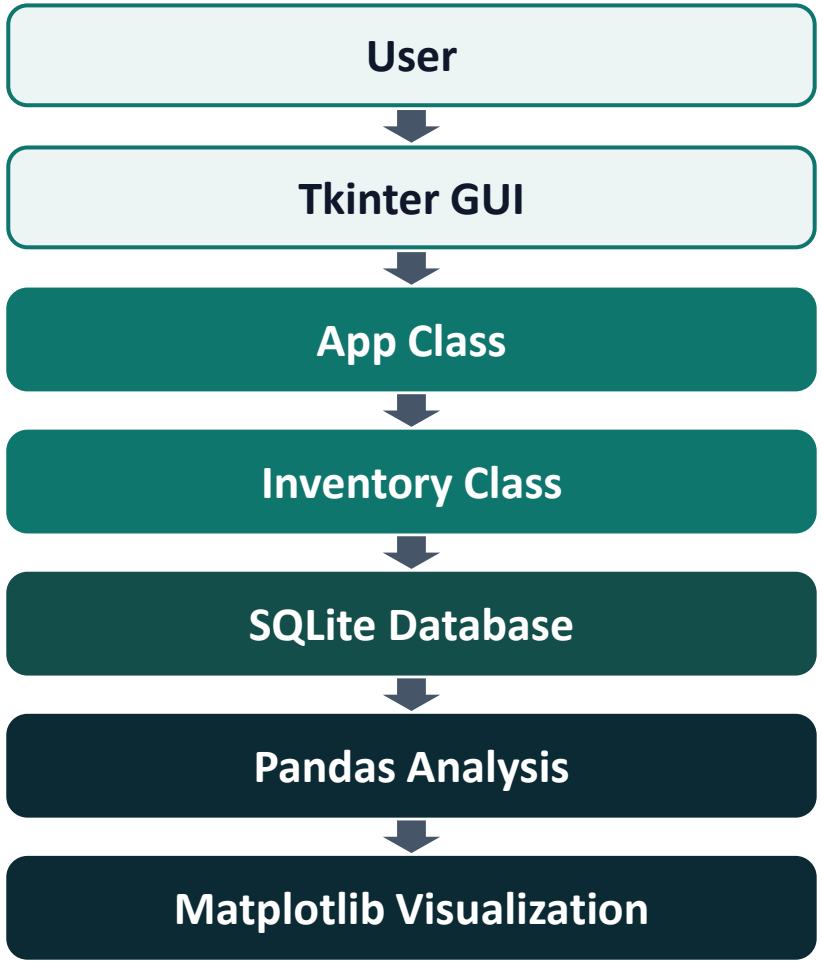
Tkinter
Graphical User Interface

SQLite
Persistent database

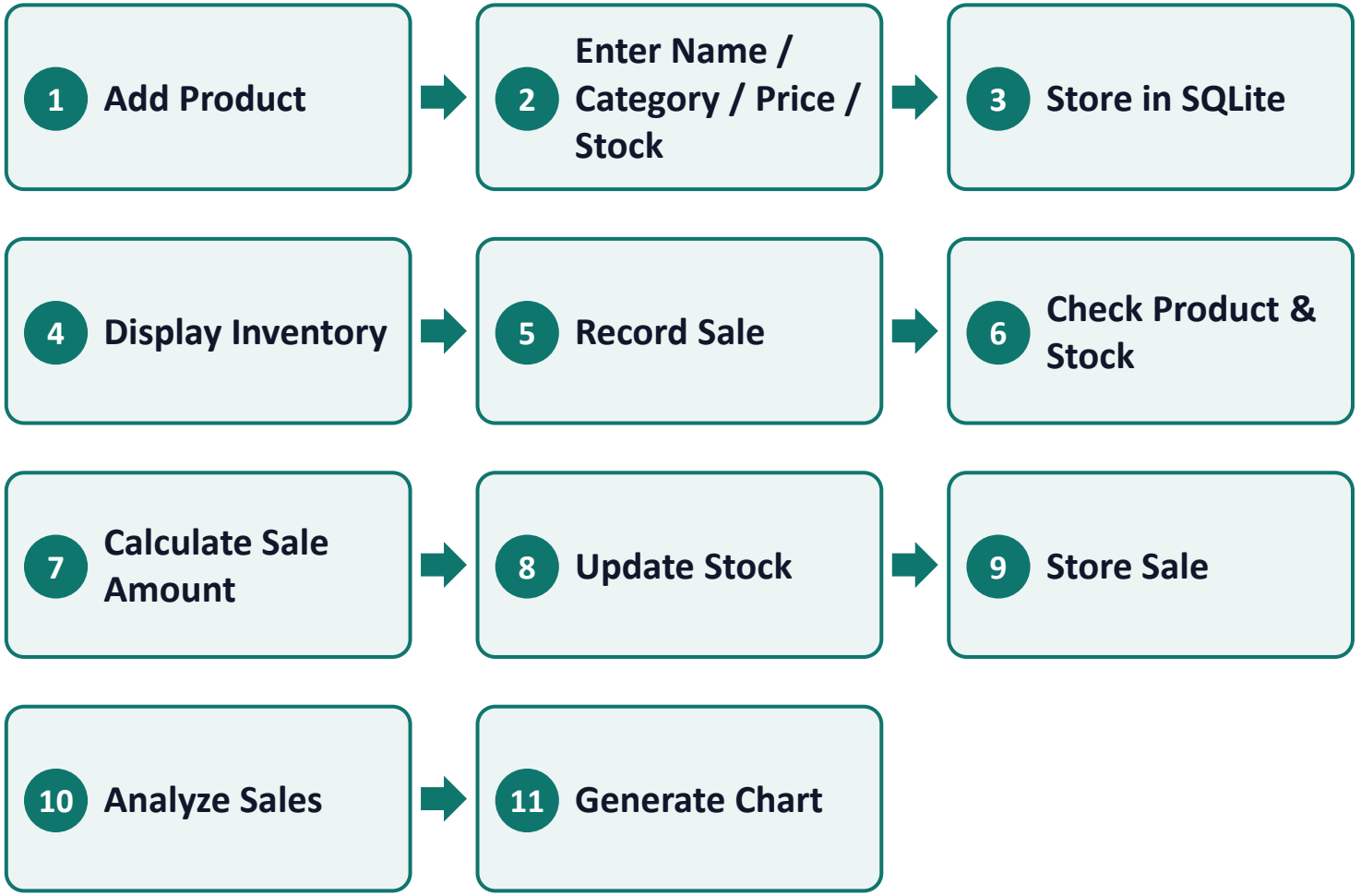
Pandas
Data analysis

Matplotlib
Data visualization

OOP
Program structure



System Design & Workflow



Inventory Class

- Database operations
- Product management
- Sales processing

App Class

- GUI creation
- User interaction
- Analytics display

Implementation & Sales Analytics

OOP

- Inventory class
- App class

SQLite

- products table
- sales table

Pandas

- DataFrame
- Total revenue
- Best-selling product
- groupby()

Matplotlib

- Category-wise sales bar chart

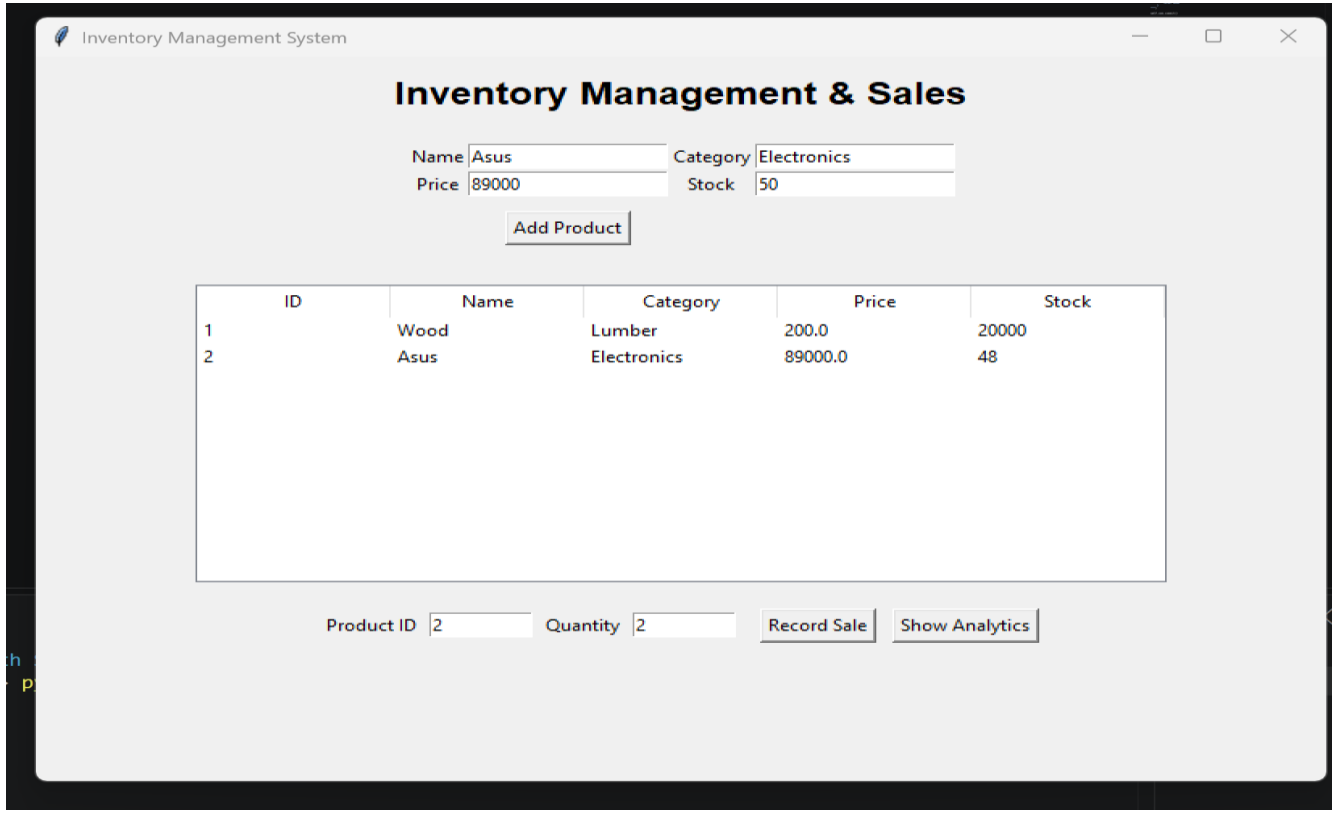
PRODUCTS table

ID	Name	Category	Price	Stock

SALES table

Product	Category	Quantity	Total

GUI Output & Results



The screenshot should show: product input fields, inventory table, Product ID and Quantity fields, Record Sale and Show Analytics buttons.

Results

- Products stored successfully
- Inventory displayed through GUI
- Sales recorded successfully
- Stock reduced automatically
- Revenue calculated
- Best-selling product identified
- Category-wise sales visualized

Conclusion & Future Scope

Conclusion

The project successfully integrates Python, OOP, Tkinter, SQLite, Pandas and Matplotlib into a practical inventory and sales management application.

The system provides:

- Inventory management
- Sales recording
- Automatic stock updates
- Basic sales analytics
- Data visualization

Future Scope

Possible future improvements, not implemented

- Product search and filtering
- Update and delete products
- Low-stock alerts
- Monthly sales reports
- CSV/PDF export
- Web-based multi-user version
- Advanced sales forecasting

THANK YOU